CRAFT Maker Space · Mary Immaculate College · companion to Code a Tiny Computer

Sort and Search, Unplugged: printables

Paper and movement activities for 5th and 6th class · Mike Borowczak, University of Central Florida

What to print

Item	Pages	Copies	Used in
Reporter sheet	2	1 per team (10 copies)	Activities 1-4
ROOT floor sign	3	1; tape to the floor at the front	Activity 4
SMALLER and LARGER signs	4-5	1 each; one on each side wall	Activity 4
Team card sets A to J	6-15	1 page per team; cut into 8	Activity 3
Number cards 1 to 31 (A5)	16-31	1 set on card stock; cut in half	Activities 2 and 4

Before the session

- Clear a floor space about 6 m by 4 m for the search tree; a hall works if the room is full.
- Give each team a Reporter sheet and a card set; keep the A5 cards for whole-class activities.
- Number cards 6 and 9 are underlined so pupils hold them the right way up.
- Each team set has eight different numbers, so no two teams can compare answers.
- Teams of three swap jobs after each activity: Runner, Tester, Reporter.

Activities inspired by CS Unplugged (csunplugged.org), which has many more sorting and searching activities.

Reporter sheet

Write the count after each activity. Swap jobs each time: Runner, Tester, Reporter.

Runner _____ **Tester** _____ **Reporter** _____

1 · Guess my number

Round 1, any guesses: how many guesses? ☐

Round 2, only “Is it bigger than ___?”: how many questions? ☐

2 · The sorting line

How many claps until the row was in order? ☐

Did any pair forget to swap? What did the Tester spot? ☐

3 · Face-down sort: one tally mark for each comparison

Strategy A: smallest first

tally marks

Total ☐

Strategy B: split and merge

tally marks

Total ☐

4 · Be a binary search tree

Tree	Number we searched for	Questions in the tree	Questions in a line
Round 1: joined at random	☐	☐	☐
Round 2: joined 1, 2, 3 …	☐	☐	☐
Round 3: joined 8, 4, 12 …	☐	☐	☐

One number on this sheet that surprised us, and why:

_____

_____

ROOT

The first pupil stands here

SMALLER

Smaller numbers go this side

LARGER

Larger numbers go this side

Team card set A · cut on the dashed lines · keep the cards face down

A

17

A

88

A

27

A

78

A

13

A

9

A

79

A

82

Team card set B · cut on the dashed lines · keep the cards face down

B

22

B

55

B

39

B

85

B

96

B

98

B

59

B

78

Team card set C · cut on the dashed lines · keep the cards face down

C

41

C

24

C

22

C

4

C

40

C

35

C

44

C

7

Team card set D · cut on the dashed lines · keep the cards face down

D

63

D

9

D

62

D

73

D

24

D

21

D

72

D

49

Team card set E · cut on the dashed lines · keep the cards face down

E

15

E

64

E

35

E

70

E

56

E

6

E

90

E

67

Team card set F · cut on the dashed lines · keep the cards face down

F

95

F

96

F

98

F

73

F

61

F

85

F

7

F

58

Team card set G · cut on the dashed lines · keep the cards face down

G

56

G

80

G

45

G

77

G

17

G

30

G

14

G

32

Team card set H · cut on the dashed lines · keep the cards face down

H

4

H

91

H

79

H

66

H

8

H

93

H

99

H

26

Team card set I · cut on the dashed lines · keep the cards face down

I

65

I

10

I

52

I

38

I

97

I

51

I

6

I

20

Team card set J · cut on the dashed lines · keep the cards face down

J

58

J

69

J

89

J

33

J

67

J

87

J

35

J

42

1

Sort and Search, Unplugged

2

Sort and Search, Unplugged

3

Sort and Search, Unplugged

4

Sort and Search, Unplugged

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29

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31

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